

Summative Test 1 · Days 5-10

MathCraft · Grade 6 Mathematics · Professor Xavier Honablue, M.Ed.

Name: _____ Date: _____ Hour: _____ Score: _____ / 100

Covers: rectangle area & two-digit multiplication (Day 5) · missing sides (Day 6) · compound rectangles (Day 7) · parallelograms, including distractor slanted sides and diagonals (Days 8-9) · multiplying decimals & fractions (Day 10).

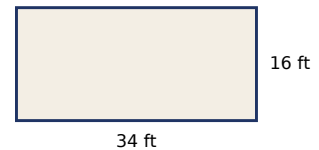
Standards: 6.G.A.1, 6.NS.B.2, 6.NS.B.3, with prerequisite 5.NF.B.4.

Directions: 25 questions · 4 points each. Show ALL work in the space provided — answers without work earn partial credit only. Label every answer with units. Figures are not drawn to scale. **Watch out:** some parallelograms show extra measurements — only base $\times$ height (the perpendicular distance) gives the area. No calculators.

Part A · Area Is Multiplication (Day 5)

1. A rectangle is 34 ft long and 16 ft wide. Find its area. (4 pts)

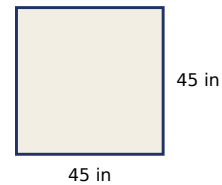
Work:



Answer: _____

2. A square rug has sides of 45 in. What is its area? (4 pts)

Work:



Answer: _____

3. **Find the mistake.** Jordan found the area of a 19 by 12 rectangle like this: $19 \times 10 = 190$, then $190 + 2 = 192$. What went wrong? What is the correct area? (4 pts)

Work:

Answer: _____

4. Which has the greater area: a 26 by 14 rectangle or a 20 by 20 square? By how much? (4 pts)

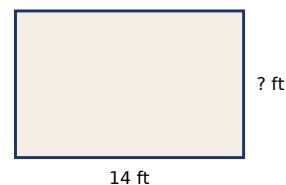
Work:

Answer: _____

Part B · Area Unlocks the Missing Side (Day 6)

5. A rectangle has an area of 126 sq ft. Its length is 14 ft. Find the width. (4 pts)

Work:



Answer: _____

6. A rectangle has an area of 552 sq m and a width of 23 m. Find the length. (4 pts)

Work:

Answer: _____

7. A square has an area of 196 sq in. Find the length of each side, then find the perimeter. (4 pts)

Work:

Answer: _____

8. Two rectangles each have an area of 48 sq units. One has a side of 4; the other has a side of 6. Find each missing side. Which rectangle has the larger perimeter? (4 pts)

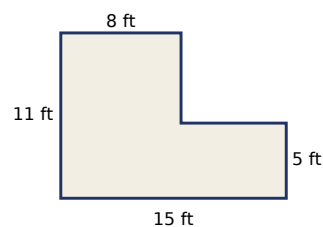
Work:

Answer: _____

Part C · Area of Compound Rectangles (Day 7)

9. Find the area of the L-shaped floor. (All angles are right angles.) (4 pts)

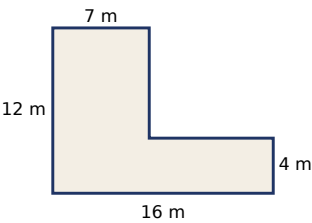
Work:



Answer: _____

10. Find the area of this L-shape TWO ways: (a) split and add, (b) enclose and subtract. Show both give the same answer. (4 pts)

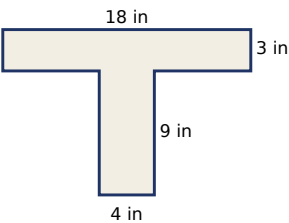
Work:



Answer: _____

11. Find the area of the T-shape. The top bar is 18 in wide and 3 in tall. The stem is 4 in wide and 9 in tall. (4 pts)

Work:



Answer: _____

12. A bedroom is 22 ft by 13 ft. A 6 ft by 3 ft closet will NOT get carpet. Carpet costs \$5 per square foot. How much will the carpet cost? (4 pts)

Work:

Answer: _____

13. Find the mistake. For the floor in Question 9, Deshawn wrote $15 \times 5 + 8 \times 11 = 75 + 88 = 163$ sq ft. What went wrong? What is the correct area? (4 pts)

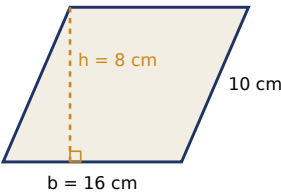
Work:

Answer: _____

Part D · Area of a Parallelogram (Days 8-9)

14. Find the area of the parallelogram. (The dashed line is the height.) (4 pts)

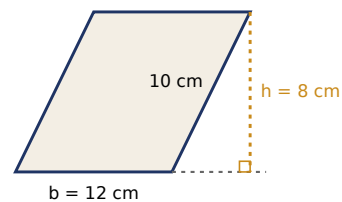
Work:



Answer: _____

15. Height outside the shape. The height of this parallelogram is measured OUTSIDE the figure: the base is extended past the slanted side (dotted), and the dotted 8 cm line drops straight down from the top corner, making a right angle with that extended base. The base is 12 cm and the slanted side is 10 cm. Find the area. (4 pts)

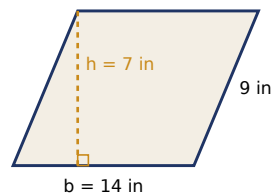
Work:



Answer: _____

16. Find the mistake. A parallelogram has base 14 in, slanted side 9 in, and height 7 in. A student said the area is $14 \times 9 = 126$ sq in. Explain the error and find the correct area. (4 pts)

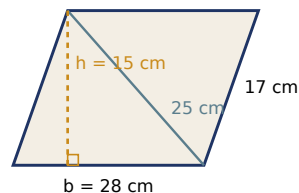
Work:



Answer: _____

17. Distractor alert. This parallelogram shows a base, a slanted side, a diagonal, and the height. Find its area. (4 pts)

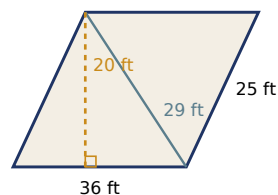
Work:



Answer: _____

18. Which one is the height? A parallelogram has a 36 ft base. It is also labeled 25 ft (a slanted side), 29 ft (a diagonal), and 20 ft (a segment that makes a right angle with the base). Name the height and find the area. (4 pts)

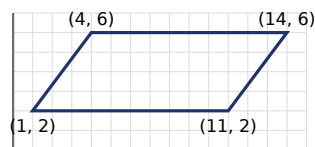
Work:



Answer: _____

19. A parallelogram has vertices (1, 2), (11, 2), (14, 6), and (4, 6). Ana says, "The side from (1, 2) to (4, 6) is 5 units long, so the height is 5." Is Ana right? Find the area. (4 pts)

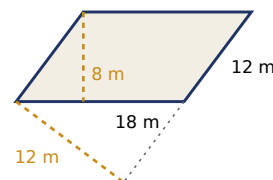
Work:



Answer: _____

20. Two bases, two heights. A parallelogram has sides 18 m and 12 m. The height to the 18 m side is 8 m, and the height to the 12 m side is 12 m. Sam wrote $18 \times 12 = 216$ sq m. What went wrong? Find the area. (4 pts)

Work:



Answer: _____

Part E · Multiplying Decimals (Day 10)

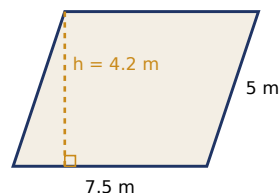
21. Multiply: 3.4×2.5 (4 pts)

Work:

Answer: _____

22. A parallelogram has a base of 7.5 m, a slanted side of 5 m, and a height of 4.2 m. Find its area. (4 pts)

Work:



Answer: _____

23. Find the mistake. Marcus says $0.5 \times 0.6 = 3.0$. Explain his error and give the correct product. (4 pts)

Work:

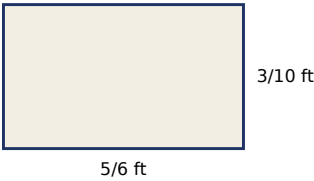
Answer: _____

Part F · Multiplying Fractions (Day 10)

24. A rectangle is $\frac{5}{6}$ ft long and $\frac{3}{10}$ ft wide. Find its area. (4 pts)

Work:

Answer: _____



25. **Challenge.** A rectangular yard is $9\frac{1}{2}$ ft by 8 ft. A shed that is 3.5 ft by 2 ft sits in the yard. How many square feet of yard are NOT covered by the shed? (4 pts)

Work:

Answer: _____

